Music Genre Classification from Spectrograms

CNN trained on Deezer previews vs. CLAP embeddings

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Abstract

A convolutional network classifies 30-second music clips into eight genres (classical, pop, jazz, hip-hop, rock, electronic, country, film score) from log-mel spectrograms. Data: 25 943 previews scraped from the public Deezer API, labelled by album genre tag, split by artist. Best from-scratch CNN: 77.4 % test accuracy on unseen artists. A small classifier on frozen CLAP audio embeddings reaches 79.4 %; zero-shot CLAP 69.4 %. Chance is 12.5 %.

1 Data

- Source: Deezer public API (no key). Spotify no longer serves audio previews.
- Per genre g with Deezer genre id: seed artists from genre charts, editorial playlists and radios; breadth-first expansion over `/artist/{id}/related`.
- Keep artist a iff $\frac{|\{\text{albums of } a \text{ with genre_id}=g\}|}{|\text{albums of } a|} \geq 0.5$. Label of a track = its album's genre id (must equal g).
- Caps: ≤ 12 tracks/artist, ≤ 4/album; artist in exactly one genre. Film score: genre "Films/Games" seeded from 65 film and game composers (genre charts skipped: pop songs from films).
- Split by artist, stratified per genre, 80/10/10. Second crawl (new artists only) doubled the training set; val/test clips unchanged.
- Fresh set: 504 clips, 258 artists absent from all splits, 63 per genre.

	clips	train	val	test
first crawl (cnn_v3)	12 699	10 214	1 281	1 204
after extension (cnn_v4)	25 943	23 458	1 281	1 204
fresh set	504	–	–	504

Table 1: Dataset sizes. ≈1 600 (later ≈3 250) clips per genre.

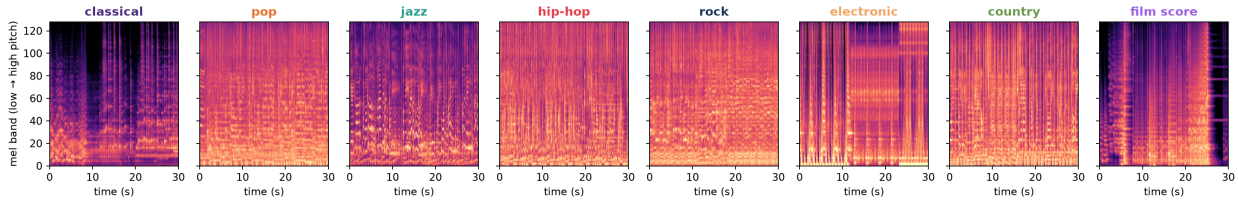


Figure 1: Most typical clip per genre (closest to the genre's mean spectrum), shared colour scale.

2 Features

Mono, resampled to $f_s = 22\,050$ Hz, 30 s $\Rightarrow x \in \mathbb{R}^{661\,500}$.

$$S[k,t] = \Big| \sum_n x[n + tH]\, w[n]\, e^{-2\pi i k n/N} \Big|^2, \quad N = 2048,\ H = 1024 \tag{1}$$

$$M[m,t] = \log\Big(\sum_k F_{m,k}\, S[k,t] + 10^{-6} \Big), \quad m = 1,\dots,128,\ t = 1,\dots,646 \tag{2}$$

F: mel filterbank, 20 Hz–11 025 Hz. Training input: random 10 s crop (216 frames) + SpecAugment (2 frequency masks ≤ 16 bins, 2 time masks $\leq 10\%$) + random gain. Evaluation: full 30 s.

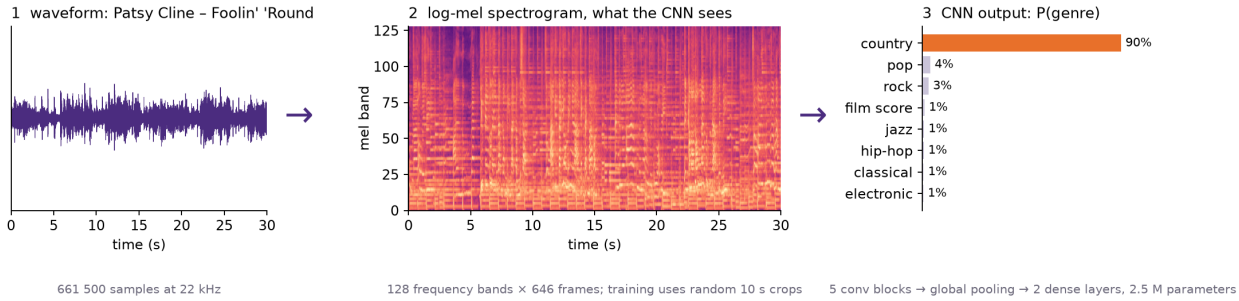


Figure 2: One test clip through the pipeline.

3 Model

layer	spec	output
input norm	BatchNorm(1)	$1 \times 128 \times T$
block $\times 5$	$2 \times$ [conv 3×3, BN, ReLU], maxpool 2	widths $w, 2w, 4w, 8w, 8w$
pooling	concat(global avg, global max)	$16w$
head	dropout 0.3, FC 256, ReLU, dropout 0.3, FC 8	8

Table 2: VGG-style CNN. $w = 32$: 2.49 M parameters (cnn_v3); $w = 48$: 5.49 M (cnn_v4).

$$\mathcal{L} = -\sum_c \tilde{y}_c \log p_c, \qquad \tilde{y}_c = (1-\epsilon)\,\mathbb{1}[c = y] + \epsilon/8, \quad \epsilon = 0.1 \tag{3}$$

$$\text{class weight}_c \propto (N/n_c)^{1/2} \tag{4}$$

AdamW, lr 10^{-3}, weight decay 10^{-2}, one-cycle schedule, batch 64. cnn_v3: $w = 32$, 40 epochs on 7 genres then 9 epochs fine-tuned with a new 8-way head. cnn_v4: $w = 48$, 11 epochs from scratch on the extended data.

4 CLAP baselines

CLAP (LAION, `laion/larger_clap_general`): audio encoder f_a and text encoder f_t into $\mathbb{R}^{512}$, trained contrastively on audio–caption pairs.

$$\mathbf{a} = \frac{1}{3}\sum_{j=1}^{3} f_a(x_j), \quad x_j = 10\,\text{s chunk } j \text{ (48 kHz)} \tag{5}$$

$$\mathbf{t}_c = \text{mean}_{s\in\mathcal{S}_c}\, f_t(s), \qquad \mathcal{S}_c = \{\text{2 descriptions} \times \text{3 templates}\} \tag{6}$$

$$p_c = \text{softmax}_c\Big(\tau\,\frac{\mathbf{a}\cdot\mathbf{t}_c}{\|\mathbf{a}\|\,\|\mathbf{t}_c\|}\Big), \qquad \tau = e^{\text{logit_scale}_a} \approx 39 \quad \text{(zero-shot)} \tag{7}$$

$$\hat{p} = \text{softmax}\left(W_2\,\text{GELU}(W_1\,\hat{\mathbf{a}})\right), \quad \hat{\mathbf{a}} = (\mathbf{a}-\mu)/\sigma,\ W_1 \in \mathbb{R}^{256\times512},\ W_2 \in \mathbb{R}^{8\times256} \quad \text{(head)} \tag{8}$$

Head: CLAP frozen; dropout 0.5, AdamW wd 0.1, 300 epochs, cosine lr; hidden size and regularisation chosen on val. Trained on the 10 214 first-crawl clips. Ensemble = mean of cnn_v4 and head probabilities.

5 Results

	params trained	test acc	test F1	fresh acc	fresh F1
cnn_v3 ($w = 32$, 10k train clips)	2.5 M	0.754	0.754	0.778	0.778
cnn_v4 ($w = 48$, 23k train clips)	5.5 M	0.774	0.773	0.794	0.794
CLAP zero-shot	0	0.694	0.696	0.728	0.728
CLAP embeddings + classifier	0.13 M	0.794	0.791	0.813	0.814
ensemble cnn_v4 + head	–	**0.797**	**0.795**	**0.815**	**0.814**

Table 3: Accuracy and macro-F1, 8 classes. Test: 1 204 clips, artists unseen in training. Fresh: 504 clips, 258 new artists. Chance = 0.125.

	test F1			fresh F1		
genre	cnn_v4	head	ens.	cnn_v4	head	ens.
classical	0.81	0.82	0.82	0.86	0.82	0.87
pop	0.62	0.65	0.67	0.59	0.69	0.62
jazz	0.80	0.81	0.82	0.84	0.83	0.85
hip-hop	0.85	0.91	0.90	0.93	0.95	0.95
rock	0.76	0.77	0.76	0.81	0.78	0.83
electronic	0.75	0.80	0.78	0.70	0.84	0.73
country	0.88	0.88	0.90	0.87	0.81	0.87
film score	0.71	0.70	0.72	0.75	0.79	0.79

Table 4: Per-genre F1.

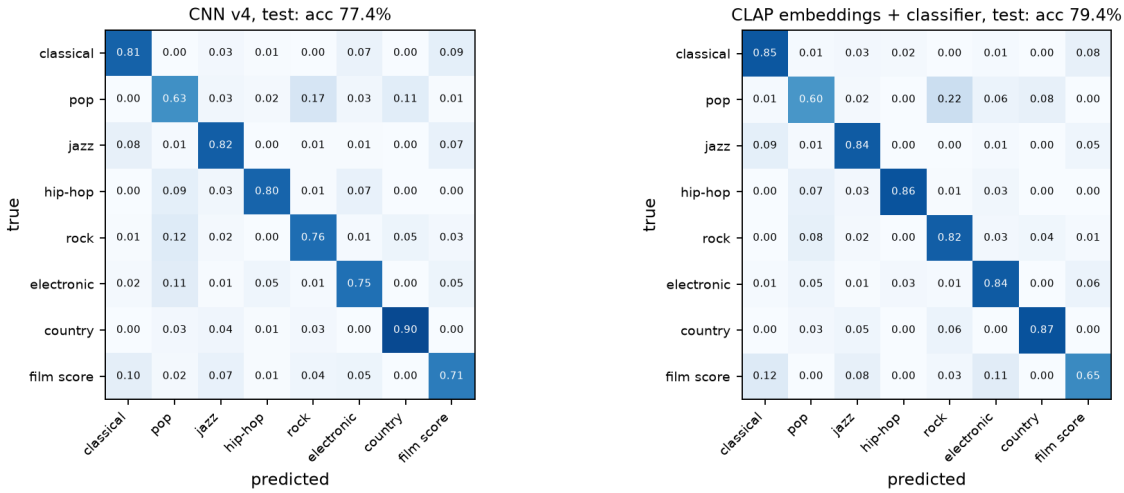


Figure 3: Row-normalised confusion matrices on the test split.

clip length	2 s	3 s	5 s	10 s	15 s	30 s
cnn_v3 test acc	0.520	0.622	0.674	0.750	0.735	0.756

Table 5: Accuracy vs. centre-cropped test clip length.

6 Observations

- Pop is the catch-all: cnn_v4 sends 11 % of pop to rock and 13 % to country; rock, hip-hop, electronic each leak 10 %–14 % into pop. Label-level ambiguity, not fixable by data.
- Classical ↔ film score: 20 % of classical predicted as score. Adding the 8th class cost classical recall 0.90 → 0.67 (cnn_v2 → cnn_v3).
- Doubling data + 2.2× parameters: +2.0 pts test, +1.6 pts fresh. Val still rising at the last epoch.
- Transfer beats scratch: 0.13 M trained parameters on frozen CLAP embeddings > 5.5 M-parameter CNN on 2× data. Ensemble adds +0.3 pts: both models fail on the same clips.
- Zero-shot CLAP calls 40/63 fresh electronic clips pop; depends on prompt wording (untuned).
- Published human agreement on broad genre labels ≈ 0.75–0.85: the models are near the label-noise ceiling.

7 Limitations and next steps

- Labels are album-level Deezer tags; 30 s previews, not full songs.
- `laion/larger_clap_music` loads with a degenerate logit scale under transformers 5; not evaluated.
- Next: tune CLAP prompts on val; longer schedule for cnn_v4; fine-tune the CLAP audio encoder end-to-end; 100k clips (≈ 1 day: 48 GB, streaming loader, $w = 64$).

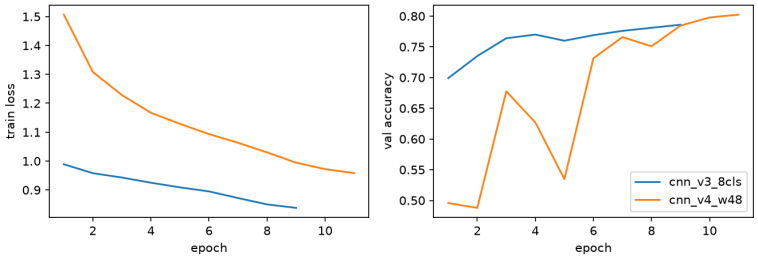


Figure 4: Training curves. cnn_v3 shown for its 9 fine-tuning epochs.

Code

`~/Desktop/College/FYS/genre/src`: `scrape_deezer.py`, `extend_data.py`, `crawl_score.py`, `preprocess.py`, `train.py`, `evaluate.py`, `clap_zero.py`, `clap_embed.py`, `clap_head.py`, `eval_all.py`, `fresh_test.py`, `app.py` (demo UI).

```
python src/scrape_deezer.py && python src/preprocess.py
python src/train.py --name cnn_v4_w48 --width 48 --epochs 11
python src/clap_embed.py && python src/clap_head.py
python src/eval_all.py cnn_v4_w48 clap_head "ensemble:cnn_v4_w48+clap_head"
```